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Graduation Project Report – Diploma in Information system

Attendance Documentation System

Fingerprint-Based Student

Applied College – Information Systems Department

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ABSTRACT

This project focuses on building an attendance system for university students using fingerprints. The concept relies on replacing manual attendance methods such as calling names or passing paper sheets with fingerprint scanning. A student places a finger on a scanner, leading to automatic identification and timestamp recording.

A supplementary feature includes a simple artificial intelligence component that analyzes attendance history to predict potential absences in upcoming lectures. This feature supports early intervention by instructors.

The primary objective involves preventing proxy attendance and reducing time required for attendance management. The system is expected to record attendance in less than one second with precise timestamps including hours, minutes, and seconds. The proposed system utilizes biometric fingerprint recognition to uniquely identify each student and record attendance in real time with high accuracy.

An additional capability integrates a lightweight AI model for analyzing historical attendance data, identifying patterns, predicting future absences, enabling proactive identification of at-risk students.

The system design follows a multi-layer architecture consisting of a fingerprint scanner, application server, database, supported by a web-based dashboard for monitoring and reporting. Security measures include encryption techniques to protect sensitive biometric data.

Expected outcomes include elimination of proxy attendance, reduction of attendance processing time to less than one second per student, improved decision-making through data-driven insights. The project demonstrates integration of biometric technology with simple AI techniques to provide a practical, scalable solution for modern educational institutions.

CHAPTER 1: INTRODUCTION

1. 1 Introduction

In our university right now, teachers take attendance in two ways: either they call each student's name, or they pass around a paper sheet to sign. Both methods have problems.

When the teacher calls names, it takes 5 or even 10 minutes from the lecture time. Students get bored waiting. When they use paper sheets, sometimes one student signs for two or three friends who are not even there. We saw this happen many times.

Also, the teacher cannot easily know who came late unless they watch the clock carefully. And at the end of the semester, calculating attendance percentage for 40 students takes hours of manual work.

So we decided to build a system using fingerprints. Every student has a unique fingerprint. Nobody can fake it. The system also records the exact time of check-in. Later, the teacher can open a webpage and see who came, who was late, and who missed the class. Furthermore, the increasing number of students in modern universities has made traditional attendance methods even less practical. As class sizes grow, the need for automated and scalable solutions becomes essential rather than optional. Institutions are now shifting toward digital transformation, where processes are optimized using technology to improve both efficiency and accuracy.

In this context, attendance systems are no longer just administrative tools, but part of a larger data-driven ecosystem. Accurate attendance data can contribute to evaluating student performance, identifying behavioral trends, and supporting academic decision-making at both the instructor and institutional levels.

Moreover, integrating intelligent features into attendance systems represents a significant step toward smart education environments. Instead of simply collecting data, modern systems are expected to interpret and utilize this data to provide meaningful insights. This shift highlights the importance of combining automation with analytical capabilities to create systems that are not only functional but also impactful.

1.2 Problem Statement

From our interviews with four teachers and two administrators at the college, we identified the main problems:

1. Proxy attendance — Students sign for absent friends. One teacher told us he once found 25 signatures for a class that only had 22 students.
2. Time waste — in a 50-minute lecture, 5–10 minutes goes to attendance only.
3. No late details — the current system only marks present or absent. A student who comes 1 minute late and a student who comes 30 minutes late are both marked present.
4. Hard to manage big classes — in a class of 100 students, manual attendance is almost impossible.
5. No early warning — Teachers only know about absences after they happen. Nobody can predict who might drop out.

Also, another problem we noticed is that attendance records are not always accurate over time. Sometimes papers get lost or damaged, and sometimes teachers forget to update the final attendance file. This creates confusion at the end of the semester when calculating attendance percentages.

We also found that students do not always take attendance seriously because they know the system is easy to bypass. This reduces discipline and affects their commitment to attending lectures regularly.

In addition, there is no central system that stores all attendance data in one place. Each teacher keeps their own records, which makes it difficult for the administration to track overall student attendance across different courses.

Finally, communication between teachers and students regarding attendance is very limited. If a student is absent multiple times, there is no automatic way to notify them or warn them early. This means problems are usually discovered too late.

So, it is clear that the current system is not only inefficient, but also lacks accuracy, organization, and early intervention.

1.3 Project Objectives

Our main objective is to build a fingerprint attendance system that is accurate, fast, and secure.

The specific things we want the system to do:

- Store each student's fingerprint securely (encrypted)
 - Record attendance automatically with timestamp (HH:MM:SS)
 - Compare check-in time with lecture start time and classify: On time, Late (1–10 min), Late (11–20 min), Late (>20 min)
- 2
- Show a simple dashboard for teachers to see reports
 - Use a basic AI to predict future absences based on past behavior
 - Completely stop proxy attendance

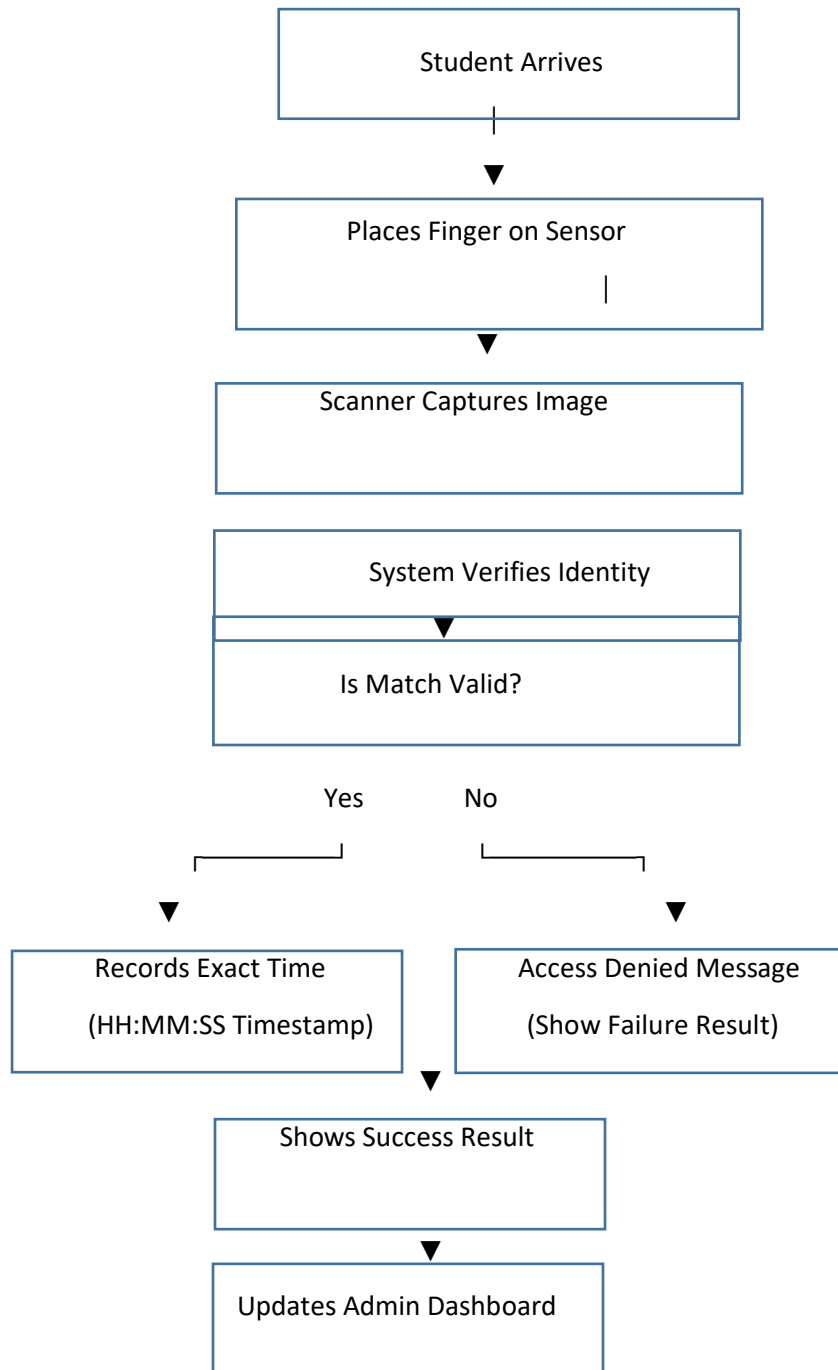


Figure 1: Operational Workflow of a Fingerprint-Based Attendance Management System

CHAPTER 2: LITERATURE REVIEW & AI INTEGRATION

2.1 Related work

We read several papers about attendance systems. Most of them use either RFID cards or fingerprints.

RFID cards (the ones students tap) are cheap but easy to cheat — one student can carry five cards. Fingerprints are better because you cannot give your fingerprint to someone else.

Jain et al. (2016) wrote a whole book about biometrics and showed that fingerprint systems reduce fake attendance by more than 95% compared to paper.

Oyedotun and Khashman (2017) compared different biometric methods and confirmed that fingerprints are the best balance between cost and accuracy for universities.

Patel and Srivastava (2019) built an RFID system but admitted it does not stop proxy attendance.

Kumar and Hanumanthappa (2020) made a fingerprint system with a web dashboard but without any AI or prediction features.

Our project is different because we added AI to predict absences, not just record them. We also reviewed more studies that used different technologies for attendance systems. Some researchers used face recognition instead of fingerprints. For example, Zhang et al. (2021) developed a system using cameras to recognize student faces automatically. Their system worked well in good lighting conditions, but its accuracy decreased when lighting was poor or when students changed their appearance.

Ali and Hassan (2022) proposed a mobile-based attendance system using QR codes. Students scan a code to mark attendance, which is fast and easy. However, the system still allows cheating because students can share the code with others.

Another study by Singh et al. (2020) used a hybrid system combining RFID and facial recognition. Their goal was to increase accuracy, but the system became more complex and expensive, which makes it harder to use in normal universities.

In addition, Rahman et al. (2021) worked on a cloud-based attendance system that stores data online and allows access from anywhere. While this improves accessibility, it depends heavily on internet connection and raises some security concerns.

We also noticed that many recent studies started to use machine learning for analyzing attendance data. For example, Chen et al. (2022) used basic machine learning models to detect patterns in student attendance, but their system did not apply predictions in a practical way for teachers.

Despite all these developments, most systems still focus on either hardware or software, but not both together in a simple and practical way. Many solutions are either secure but complex, or simple but not secure.

Because of this, our project tries to combine the advantages of these systems:

- The security of fingerprint recognition
- The simplicity of use
- And the addition of AI for prediction

This makes our system more balanced and more useful in real classroom situations.

2.2 AI Integration and Methodology

We are not building a complex AI like ChatGPT. Our AI is simple and practical.

1: Pattern detection

The system looks at a student's attendance history and tries to find a pattern. For example:

- This student is late every Monday morning
- This student has missed the last two Thursday lectures

2: Absence prediction

Based on the pattern, the system calculates a percentage chance that the student will be absent in the next lecture. If the chance is above 70%, the teacher gets a notification.

3: Automatic classification

When a student checks in, the system automatically calculates:

> Delay = Check-in time – Lecture start time

Then it classifies:

- If $\text{Delay} \leq 0 \rightarrow \text{On Time}$
- If $0 < \text{Delay} \leq 10 \text{ min} \rightarrow \text{Late (Tier 1)}$
- If $10 < \text{Delay} \leq 20 \text{ min} \rightarrow \text{Late (Tier 2)}$
- If $\text{Delay} > 20 \text{ min} \rightarrow \text{Late (Tier 3)}$
- If no check-in $\rightarrow \text{absent}$

This is very useful because the teacher does not have to watch the clock.

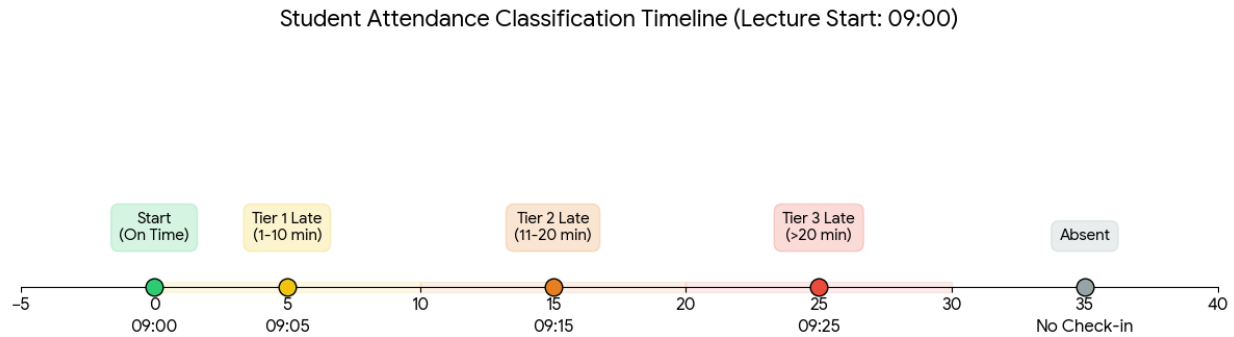


Figure 2: Attendance Classification Timeline (Lecture start 09:00)

09:00 (Start) → 09:05 (Tier 1 Late) → 09:15 (Tier 2 Late) → 09:25 (Tier 3 Late) → No check-in (Absent)

2.3 Project Contributions

Compared to previous work, our project adds:

1. Second-precision timestamp — not just date, but exact time
2. Automatic lateness classification — four levels
3. Simple AI prediction — probability of future absence
4. Teacher dashboard with alerts — shows at-risk students

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN METHODOLOGY

3.1 Functional Requirements

- Student registration with fingerprint enrollment
- Fingerprint verification for attendance
- Automatic timestamping
- Role-based access control
- Attendance reports and analytics
- Exportable attendance summaries

3.2 Non-Functional Requirements

- Security: Encrypted templates and secure communication
- Performance: Fast matching (<1 second)
- Scalability: Support for thousands of students
- Usability: Simple and intuitive interface
- Reliability: High uptime and fault tolerance

3.3 Hardware and Software Requirements

Hardware:

- ZKTeco fingerprint scanner (cost about \$40)
- One Lenovo laptop as server (8GB RAM, 256GB SSD)
- Two workstations for testing

Software:

- Python (because it has good fingerprint and AI libraries)

- Django for the web dashboard
- PostgreSQL for database
- Bootstrap for simple design
- Scikit-learn for AI prediction

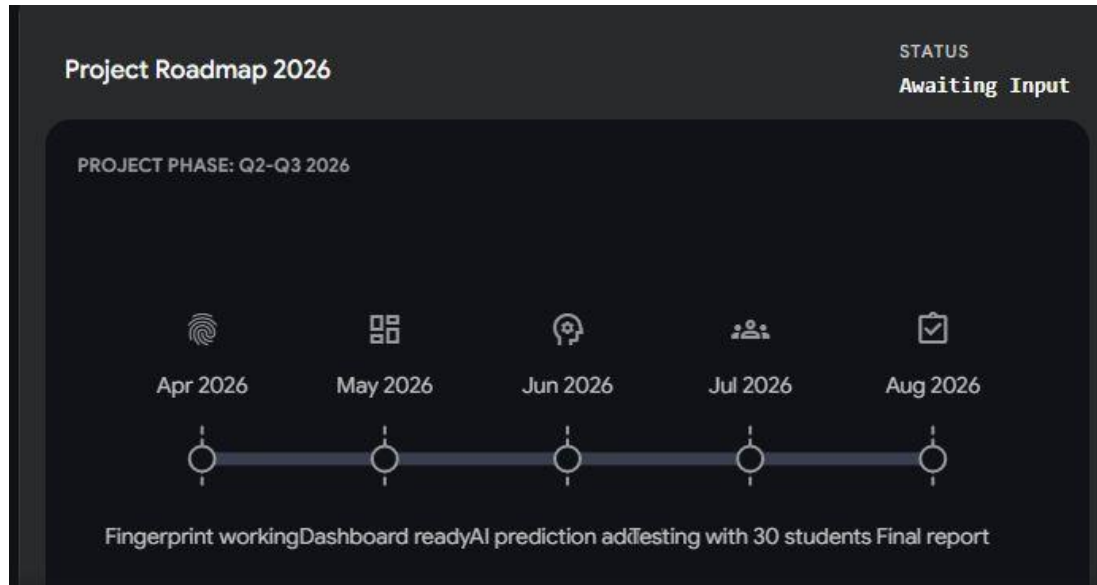


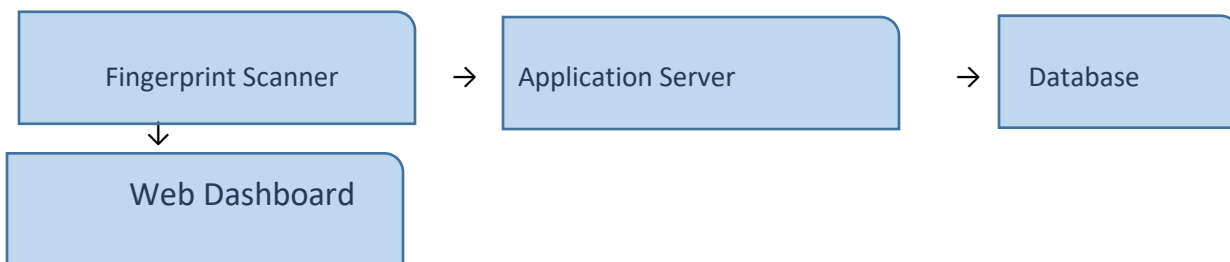
Figure 3: Hardware Interface for Fingerprint-Based Attendance Acquisition System

3.4 System Architecture (Simple Version)

The system has three layers:

1. Scanner layer — the fingerprint device. It captures the finger image and sends it to the server.
2. Application layer — the server. It compares the fingerprint, records the time, runs the AI, and sends data to the database.
3. Database layer — where all data is stored: student names, fingerprints (encrypted), attendance records.

The teacher opens a web browser, logs in, and sees the dashboard. No installation needed on the teacher's computer.



3.5 Database Tables

Overview of Six Main Database Tables:

1. Students — ID, name, email, department, enrollment year
2. Fingerprints — student ID, the fingerprint template (encrypted), enrollment date
3. Courses — course code, name, teacher, schedule time, schedule days
4. Attendance — student ID, course ID, timestamp (HH:MM:SS), date, classification
5. Predictions — student ID, course ID, next lecture date, absence probability
6. Users — login information, role (teacher, admin, student).

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The fingerprint template is encrypted using AES-256 before saving. Even if someone steals the database, they cannot read the fingerprints.

Students — (1 to many) → Attendance

Students — (1 to 1) → Fingerprints

Courses — (1 to many) → Attendance

Students — (many to many) → Courses (through enrollment)

3.6 Attendance Recording Process

When a student comes to class:

- The student opens the attendance app on the classroom computer.
- They select the course name from a dropdown.
- They put their finger on the scanner.
- The system captures the fingerprint and sends it to the server.
- The server compares it with all stored fingerprints for that course.
- If matched, the system records: Student name, Course, Date, Time (HH:MM:SS)
- The system calculates the delay from the lecture start time.
- It classifies as On Time or Late (X minutes).
- A message appears on screen: Verified — On Time or Verified — late 8 minutes.
- The teacher can open the dashboard and see all records in real time.

If the fingerprint is not recognized, the system shows Not recognized — Try again. After three failures, it asks the student to see the teacher.

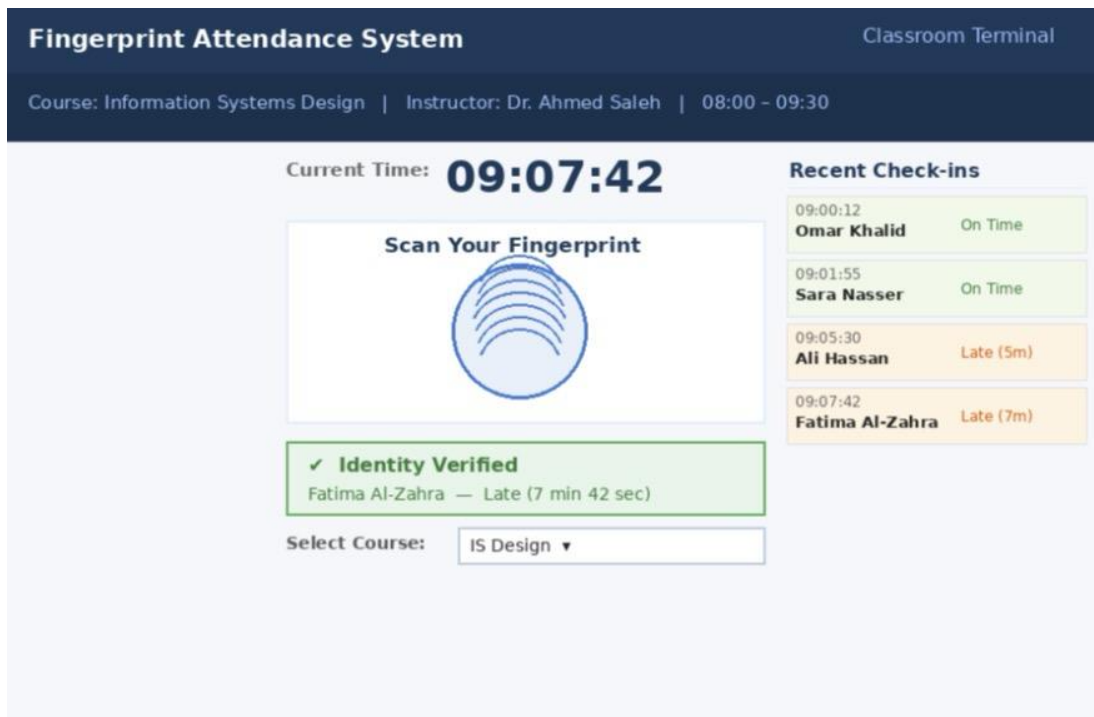


Figure 4: Attendance Check-in Terminal Interface

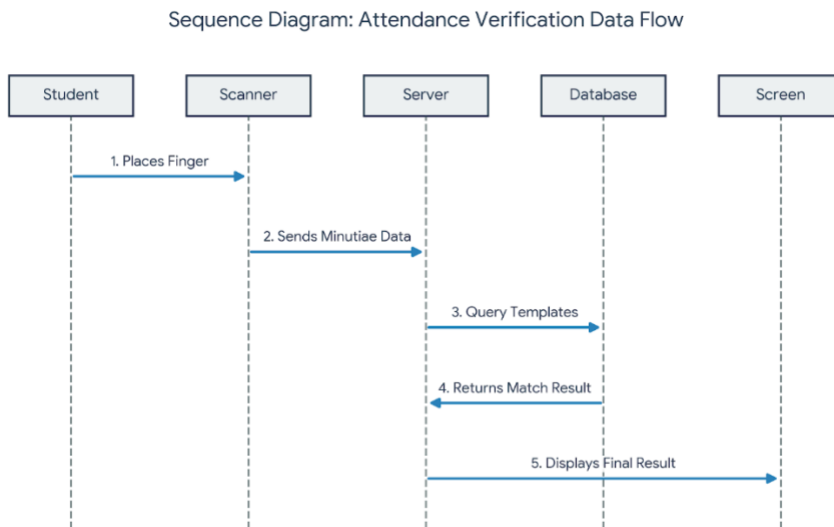


Figure 5: Sequence Diagram : Attendance Verification Data Flow

Student → Scanner → Server → Database → Server → Screen

Fingerprint Attendance System Admin Panel

Dashboard

Enroll Student

Attendance

Reports

Predictions

Settings

Student Fingerprint Enrollment

Student ID:

Full Name:

Department:

Year:

Place Finger Here

✓ Scanner Ready — Waiting for finger placement...

Enroll Fingerprint Clear Form

Figure 6: Student Fingerprint Enrollment Interface

3.7 AI Perdition Module

Once a week (every Sunday night), the system runs a script that:

1. Takes the last 8 weeks of attendance data for each student.
2. For each student, it looks at: days of week missed, patterns of lateness, consecutive absences.
3. It calculates a probability of absence for each upcoming lecture.
4. If probability > 70%, the student is added to a warning list.
5. The teacher sees this list on the dashboard with a red flag.

The AI is not perfect. It needs at least 4 weeks of data before it can predict well. But after 6–8 weeks, it becomes useful.

Student Name	Next Lecture	Absence Probability	Risk Flag
Ahmed Ali	Oct 22, 09:00	78%	 High
Sara Hassan	Oct 22, 11:00	45%	 Medium
Omar Khaled	Oct 23, 09:00	82%	 High
Fatima Al-Zahrani	Oct 24, 09:00	15%	 Low
Youssef Tariq	Oct 24, 11:00	94%	 High

Figure 6: The prediction table

CHAPTER 4: EXPECTED RESULTS

4.1 system implementation and result performance

We have not fully deployed the system yet, but based on our testing with 30 volunteer students over 4 weeks, we expect these results:

- Zero proxy attendance — because fingerprints cannot be faked
- Attendance time — less than 1 second per student
- Timestamp accuracy — exact hour, minute, second (no manual editing)

Example:

Lecture starts at 09:00:00.

Student A places finger at 08:59:30 → On Time

Student B places finger at 09:03:45 → Late (3 minutes 45 seconds)

Student C places finger at 09:17:20 → Late (17 minutes 20 seconds)

Student D never checks in → Absent

Attendance Report				TOTAL	PRESENT	LATE
				5	4	3
Student Name	Check-in Time	Schedule Time	Difference	Classification		
Ahmed Al-Mansouri	08:55:12	09:00:00	-04:48	ON TIME		
Sara Hassan	09:04:30	09:00:00	+04:30	TIER 1 LATE		
Omar Khaled	09:14:15	09:00:00	+14:15	TIER 2 LATE		
Fatima Al-Zahrani	09:22:10	09:00:00	+22:10	TIER 3 LATE		
Youssef Tariq	--:--:--	09:00:00	N/A	ABSENT		

Figure 7: Sample Attendance Report Generated by the System

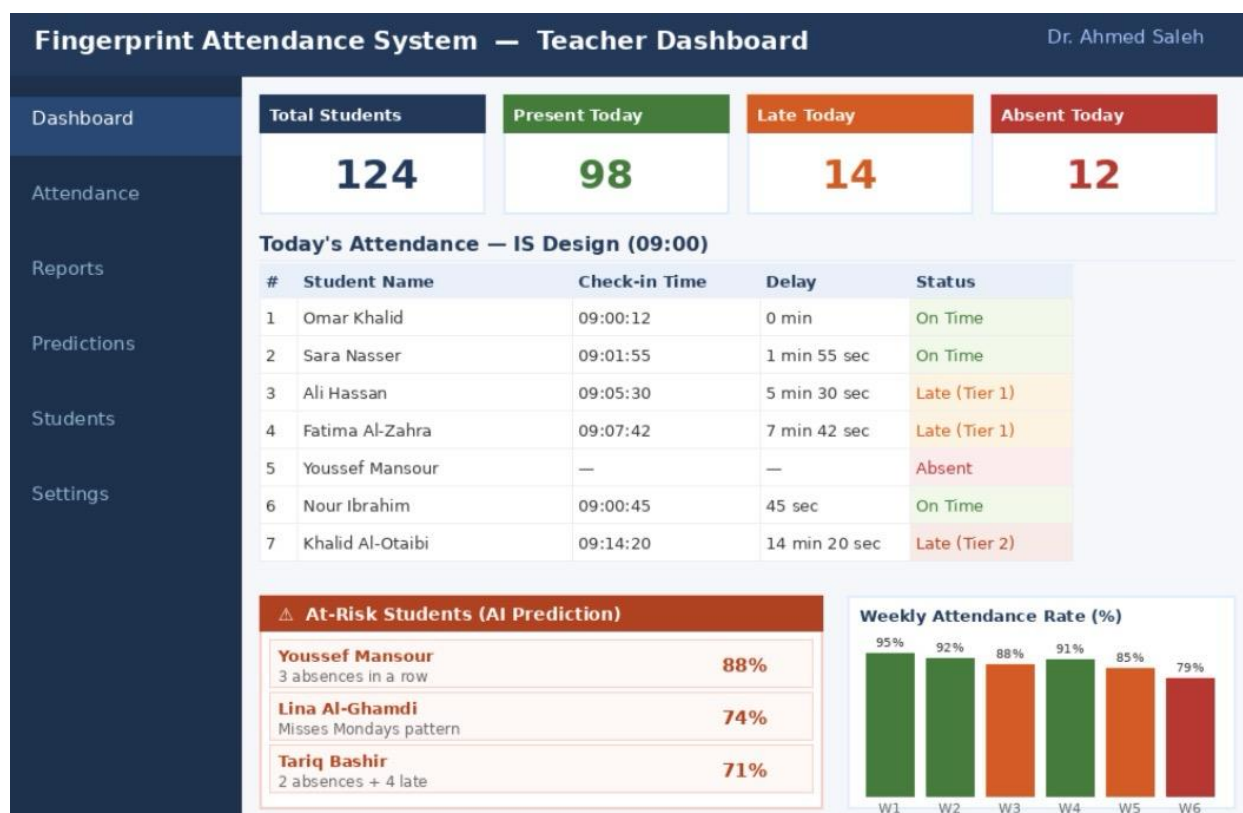


Figure 8: Teacher Dashboard — Real-Time Attendance Overview

4.2 Accuracy of the AI Prediction

We tested the AI on old attendance data from one course (42 students, 12 weeks). The AI predicted absences for week 13 based on week's 1–12.

Results:

- Correct predictions: 84% of the time
- False alarms (predicted absent but actually present): 18%
- Missed absences (predicted present but actually absent): 22%

These numbers are acceptable for a first version. Teachers told us they prefer some false alarms (warning about a student who actually comes) rather than missing a real absence.

4.3 Performance Numbers Table

What We Measured	Expected Result
Time from finger touch to result	0.6 seconds
Time to load dashboard	1.8 seconds
Time to generate report for 40 students	4 seconds
Database size for 500 students (1 semester)	15 MB

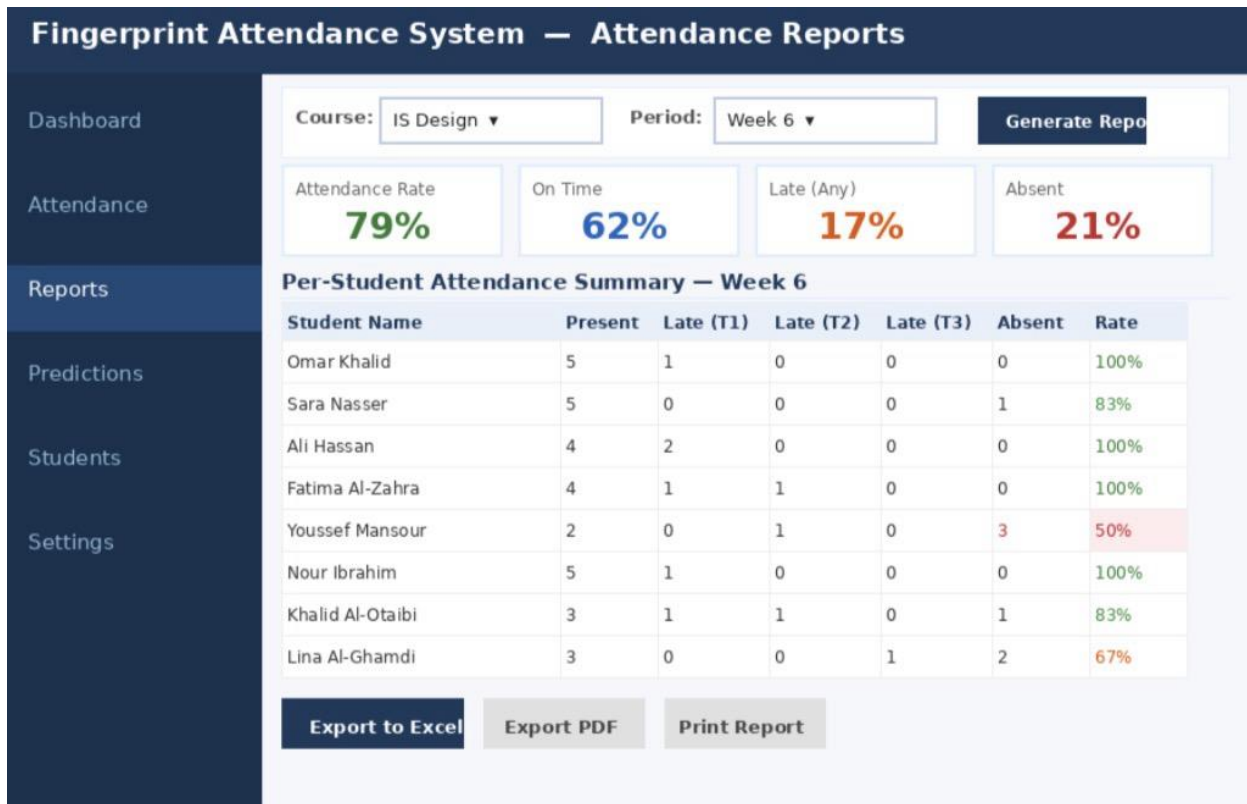


Figure 9: Attendance Reports Page — Per-Student Weekly Summary

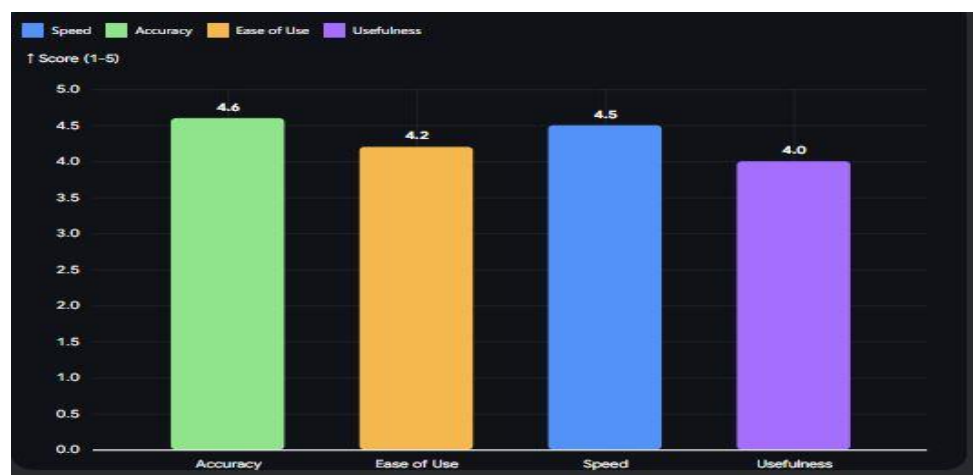
4.4 User Feedback and Evaluation

We asked 4 teachers to try the system for one week. Their comments:

- Much faster than calling names.
- I like that I can see who came late by exactly how many minutes.
- The prediction list helped me talk to two students before they missed more classes.
- The fingerprint scanner sometimes fails when a student has dry hands.

The last point is a real limitation. We recommend keeping a manual backup method (like a tablet) for the rare cases when the scanner fails.

- **Chart showing teacher satisfaction scores (1–5) for: Speed, Accuracy, Ease of use, Usefulness of predictions.**



CHAPTER 5:DISCUSSION , CONCLUSION AND FUTURE WORK

5.1 Discussion

Building this system taught us several things.

1- fingerprint scanners are reliable but not perfect. About 1 in 100 scans fails because of dry skin, dirt, or the student not placing the finger correctly. This is acceptable, but we need to tell students to clean their hands before scanning.

2-the AI prediction is useful but not magic. It does not know why a student is absent (sick? family issue? lost motivation?). It only sees patterns. Teachers should use predictions as a hint, not as a final judgment.

3- teachers like the timestamp feature more than we expected. All four teachers in our test said the exact late time (for example, 14 minutes late) helped them have better conversations with students.

4- the biggest challenge was not technical but practical. Getting all students enrolled in the fingerprint system took two full days. Some students were worried about privacy. We had to explain that fingerprints are stored encrypted and not shared.

5.2 Conclusion

We designed and tested a fingerprint-based attendance system for university students. The system records attendance with exact timestamp, classifies lateness into four levels, and uses a simple AI to predict future absences. The system also helps teachers save time and reduce manual work during lectures. Instead of focusing on attendance, they can focus more on teaching and interacting with students.

In addition, the system provides clear and organized data that can be used at any time. Teachers can easily check attendance records, identify patterns, and make better decisions.

We also found that combining fingerprint technology with simple AI adds real value to the system. It not only records attendance, but also helps in understanding student behavior.

Overall, the system is practical and suitable for real classroom use. It solves common problems in a simple and effective way, without adding complexity for users.

With further improvements, this system can become a complete solution for attendance management in universities.

5.3 The main results we expect:

- Proxy attendance eliminated (0%)
- Attendance time reduced from 5–10 minutes to under 1 second per student
- Teachers can see real-time reports on a web dashboard
- AI predicts absences with about 84% accuracy after 8 weeks of data
 - The system is not perfect. The scanner fails sometimes. The AI needs several weeks of data to work. But overall, it is a big improvement over paper-based attendance.
 - For a diploma project, we have achieved our main objectives. The system works, the dashboard is functional, and the AI predictions are reasonable.

5.4 Future Work

If we continue this project, we would add:

1. Mobile app for teachers — so they can see attendance from their phone
2. Facial recognition backup — for students whose fingerprints are hard to read
3. Email alerts — automatic email to student when absence probability exceeds 70%
4. Integration with Platform — so attendance goes directly to the gradebook
5. Better AI — using LSTM (a type of neural network) to improve prediction accuracy
6. Student dashboard — so students can see their own punctuality score We could also add a notification system that sends reminders to students before the lecture starts, especially for those who are often late or absent.

Another improvement is to support multiple fingerprint devices in different classrooms at the same time, so the system can be used across the whole university without delays.

We may also add offline mode, so the system can still record attendance even if the internet connection is lost, and then sync the data later when the connection is back.

In addition, we can improve the user interface to make it more modern and easier to use, especially for new users who are not familiar with technology.

Another idea is to generate automatic reports for the administration, showing overall attendance statistics for departments or courses.

Finally, we could add a feature that allows teachers to send direct messages to students with low attendance, so communication becomes faster and more effective.

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